



Potential, optimization and sensitivity analysis of photovoltaic-diesel-battery hybrid energy system for rural electrification in Algeria



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ABSTRACT

Integration of hybrid renewable energy systems (HRES) as an electrification solution can enhance the rural electrification situation in Algeria's predominantly remote Saharan regions, where diesel generators are used to provide very basic and limited electricity service. The exploitation of such a solution requires a sustainable, optimized HRES design. This paper presents a methodology both to optimize and to perform a sensitivity analysis of an autonomous hybrid PV-diesel-battery energy system. Particle Swarm Optimization (PSO) and ϵ -constraint method were used to simultaneously minimize total system cost, unmet load, and CO₂ emissions. One optimal solution was chosen among the solutions set and analyzed. The remote Saharan village of Tiberkatine, province of Tamanrasset in southern Algeria was considered as a case study. The hybrid system supplied the energy demand of 20 households. It was found that the optimal solution can supply energy without unmet load, the minimum cost of energy COE is 0.37 \$/kWh with 93% renewable fraction. A sensitivity analysis was performed on the optimal system to study the impact of three parameters (load consumption, ϵ_{CO_2} , and ϵ_{LLP} constraints) on the system behavior. Finally, a comparative analysis between PSO and HOMER was performed, it was found that the PSO based approach is more cost effective with more PV penetration than HOMER.

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1. Introduction

The diesel generator is the prevailing energy source in rural and Saharan regions in Algeria, but this electrification solution suffers from inefficiency when run at low load, high operation and maintenance cost associated with serious environmental problems such as GHG emissions. Decentralized generation units based on local renewable resources such as PV and wind standalone systems can be the cleanest solution to supply electricity to isolated and scattered areas, given that solar energy has the most potential among the different renewable energy sources in Algeria (the average solar energy received is 2400 kWh/m²/year, ranging from 1700 kWh/m²/year for the coastal region, 1900 kWh/m²/year for the high plateau, and 2650 kWh/m²/year for the south [1]). However, the standalone systems have various disadvantages, such as their high cost and their over-dependence on environmental conditions, which varies

from place to place, therefore, these disadvantages can lead to designing flaws [2]. In order to provide access to electricity in rural and remote regions in Algeria, Hybridization the available renewable resources with diesel generator will make an effective solution to increase system reliability.

An off-grid hybrid renewable energy system (HRES) is a combination between two or more renewable energy sources or at least one renewable source to conventional source. The HRES can be a viable alternative and the most effective solution which keeps into account the advantages of different sources of energy. Several papers have reviewed the power generation and reliability of standalone HRES. Perera et al. [3] carried out a multi-objective optimization design of HRES in order to support decision-making in the integration of renewable energy sources at early design stages of the electrification project. Domenech et al. [4] proposed a mathematical model to optimize the design of wind-photovoltaic projects by minimizing the system cost and meeting the technical, the management and the security of the supply constraints. Saheb Koussa et al. [5] investigated the performance of an experimental Hybrid diesel-wind battery system operating in standalone

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Nomenclature			
ACC	annual capital cost [\$]	N_{bat}	Number of batteries
ACS	annual cost of the system [\$]	NOCT	Nominal operation cell temperature [$^{\circ}\text{C}$]
AFC	annual fuel cost [\$]	N_{pv}	Number of modules
AOM	annual operation and maintenance cost [\$]	P_{a-DG}	Power output of diesel generator [kW]
ARC	annual replacement cost [\$]	P_{best}	Individual best position
$C_{ann,tot}$	total annualized cost [\$]	P_{ch}	Hourly demand [W]
C_1	acceleration factor	P_{in}	Power transiting the inverter [W]
C_2	acceleration factor	P_j	capacity of the component [kW]
C_B	availability power of battery [Ah]	P_{n-DG}	Nominal power of diesel generator [kW]
C_{Bmax}	maximum allowable energy level remained in battery [Ah]	P_{pv}	Output power of PV generator [W]
C_{Bmin}	minimum allowable energy level remained in battery [Ah]	PSO	Particle swarm optimization
C_{cap}	capital cost [\$]	P_{STC}	Output power of the PV generator in STC [Wp]
$CO_{2emission}$	system CO_2 emission [kg/year]	R_1	Random number
COE	levelized cost of energy [\$/kW]	R_2	Random number
$Cons_G$	consumption of diesel generator [l/h]	R_{proj}	Project life time [years]
C_{rep}	replacement cost [\$]	SFF	Sinking fund factor
CRF	capital recovery factor	SOC	Battery state of charge [%]
$D(t)$	electricity demand [kWh]	SOC_{max}	SOC upper limit [%]
$E_{(grid,sales)}$	amount of energy sold to the grid per year [kW]	SOC_{min}	SOC lower limit [%]
E_{def}	amount of deferrable load per year [kW]	$Ta(t)$	ambient temperature [$^{\circ}\text{C}$]
EF	emission factor [kg/l]	T_c	PV cell temperature [$^{\circ}\text{C}$]
E_j	energy produced or entered to each component [kWh]	v	Velocity
E_{prim}	amount of primary load per year [kW]	x	position
F_{dirt}	dirt losses factor	y	project lifetime
$fuel_{cons}$	fuel consumption of diesel generator	y_{rep}	lifetime of battery bank [years]
g_{best}	global best position	Greek symbols	
G_h	irradiation over the tilted surface of the PV generator [kWh/m ²]	α	power temperature coefficient [%/ $^{\circ}\text{C}$]
GHG	greenhouse gases	α_{DG}	coefficient of the consumption curve of DG [l/kWh]
HRES	hybrid renewable energy system	β_{DG}	coefficient of the consumption curve of DG [l/kWh]
i	annual real interest rate	ϵ_{LLP}	LLP desirable level that must not be exceeded [%]
j	iteration number	ϵ_{CO2}	CO_2 emission desirable level that must not be exceeded [kg/year]
K	constriction factor	η_G	Diesel generator efficiency
LHV_{fuel}	fuel heating value [kWh/kg]	η_{inv}	Inverter coefficient [%]
LLP	loss of load probability [%]	λ	Reliability of component
		σ	Self-discharge rate of battery
		ω	Inertia weight
		ω_0	initial weight

mode. The system was evaluated under three scenarios of state of charge of the batteries and wind speed. Ma et al. [6] have evaluated the long term operation performance of a 19.8 kWp standalone solar photovoltaic system in a remote island of Hong Kong. The methodology was based on real meteorological conditions through long-term monitoring and assessment of the system. Various research works have confirmed that HRES in off-grid applications are economically feasible, mainly in remote and rural locations. Rohani and Nour [7] evaluated the performance of different combination of PV, wind turbines, batteries and generators in order to find the optimal hybrid system to fulfill the primary load for 250, 500 and 2500 households in remote area based on lower net present cost method. Bortolini et al. [8] proposed a bi-objective design model for off-grid PV–battery–Diesel generator hybrid energy system. The aim is to identify the PV plant rated power, the battery bank capacity and the technical configuration. The levelised cost of the electricity (LCOE) and the carbon footprint of energy (CFOE) were considered as objective functions. Furthermore, climate can make one energy source of hybrid system more profitable than another source, for example, hybrid PV–diesel system with battery bank is ideal in areas with warm climates [9], which is the hybrid

system considered in our study.

To explore the advantages of HRES, we should study the ways to increase their efficiency and guarantee the durability on human scale mostly in small and remote villages where they need energy to improve the standards of their life. Optimally sized design of hybrid renewable energy system (HRES) is an important issue in hybrid systems [10], therefore, many researchers analyzed and evaluated strategies, control, configurations, methodologies and criteria for optimization of the Hybrid Renewable Energy System (HRES): Upadhyay & Sharma [2] reviewed in detail the different configurations, control and sizing methodologies of hybrid energy systems. Siddaiah and Saini [11] examined the planning, configurations, modeling and optimization techniques of hybrid renewable energy systems for off grid applications. Moreover, the selection of evaluation criteria is one of the important phases necessary for designing HES [12], Rubio et al. [13] explained in detail various criteria considered in sizing a hybrid energy system. Further, software tools such as HOMER, RETSCREEN and HOGA, are also widely used to optimize and size the HRES. Salehin et al. [14] employed both HOMER and RETSCREEN to find the modeling framework for assigning Solar PV–Diesel energy system and Wind–Diesel energy.

Mirzahosseini and Taheri [15] proposed a design model using RETSCREEN and based on electricity tariffs. Three scenarios were developed according to the targeting of energy subsidies. In Ref. [16], Carroquino et al. proposed a cost optimization design for off-grid hybrid photovoltaic-diesel systems. HOGA (Hybrid Optimization by Genetic Algorithms) software was applied to find the optimal design to power drip irrigation facilities. Based on the literature review, several research papers have been carried out to find the optimum design of HRES using different optimization methods such as analytical, numerical, probabilistic and heuristic methods. Nadjemi et al. [17] developed a multi-objective approach based on cuckoo search algorithm for grid-connected hybrid PV wind energy system. The study considered two objective functions, the total cost and grid power absorption probability (GPAP) parameter. Dufo-López et al. [18] proposed stochastic-heuristic model to minimize the net present cost of stand-alone hybrid PV-wind-diesel with battery storage systems. The stochastic optimization is developed by means of Monte Carlo simulation. The heuristic approach uses genetic algorithms to obtain the optimal system. Ibrahim et al. [19] proposed an optimization model to minimize the loss of load probability (LLP) of hybrid PV battery system. Different system's configurations with different availability levels are generated using a numerical algorithm. Recently, instead of using traditional optimization methods, heuristic methods, mainly population-based meta-heuristics such as genetic algorithms (GA), evolutionary algorithms (EA), scatter search (SS), ant colony optimization (ACO), particle swarm optimization (PSO), artificial bee colony optimization (ABCO), etc. are broadly used to solve HRES optimization problems. PSO (Particle Swarm Optimization) is one of these heuristic methods, which has been effectively applied to solve large range of problems of HRES [20] like planning, designing and control. Finding optima of complex search processes through iteration of each particle of population is the basic of PSO method [21]. PSO has few parameters to adjust, is easy to implement and has special characteristic of memory [20]. Wang and Singh [22] proposed multiple criteria approach including cost, reliability, and emissions to find the optimal design of hybrid PV-wind-battery system. An improved particle swarm optimization algorithm was employed under different design scenarios. Hakimi and Moghaddas-Tafreshi [23] used PSO algorithm to minimize the total costs of the be stand-alone hybrid power system. The system consists of wind, fuel cells, an electrolyser, a reformer, an anaerobic reactor and a hydrogen tank. Biomass is also considered in the system as an energy resource. Tudu et al. [24] presented a comparative analysis between PSO and GA shows that the PSO technique performs better than GA, when applied for a sizing problem in terms of number of iterations and CPU utilization time. Sharafi & ELMekkawy [25] proposed PSO simulation based approach to optimize and design a renewable hybrid system including various generators and storage devices, the method has been applied to minimize simultaneously the total cost of the system, CO₂ emission and unmet load. There were more than 20 varieties in the studies that used PSO algorithms [26] such as the Binary Particle Swarm optimization (BPSO), Hybrid Particle Swarm Optimization and Wavelet Mutation (HPSOWM) [27], etc. Maleki et al. [28] studied the performance of different PSO variants (PSO, modified PSO (MPSO), PSO based on repulsion factor (PSO-RF), PSO with constriction factor (PSO-CF), and PSO with adaptive inertia weight (PSO-W)) to determine the optimum sizing of hybrid (PV-wind- Battery) system.

Design, optimization and implementation of hybrid renewable energy systems is complex problem usually require multi-objective methods to solve it. Multi-objective evolutionary algorithms (MOEA), which classified in population-based methods [29] are

suitable because they have the ability to attain the global optimum [30]. According to Mezura-Montes et al. [31], multi-objective algorithms are typically divided into 3 classes: (1) Pareto-based techniques deal with multiple objectives using ranking and selection in the population. (2) Non-Pareto-Based techniques deal with combination of objective functions and problem transformation. (3) Mixed techniques that use both approaches. In recent research works for optimization of a renewable energy unit, there is an increase in usage of evolutionary computations, due to their suitability for multi-objective issues by implementing a heuristic algorithm. Tezer et al. [32] reviewed the state-of-art of hybrid system sizing approaches from past to present to determine the efficacy of multi-objective optimization approaches. Al-falahi et al. [33] examined the recent developments in size optimization methodologies, and compared single algorithms, hybrid algorithms, and software tools used for sizing standalone solar and wind HRES. Cui et al. [34] reviewed multi-objective optimization methods application in the environmental protection fields, for optimization objectives of energy saving, emissions reduction and cost reduction, etc.

The aim of this paper is to optimize and evaluate the potential of the hybrid PV-diesel-battery energy system as a reliable and sustainable rural electrification solution. PSO based approach and ϵ -constraint method (MOP technique) is used in the optimization process, the total system cost, the unmet load, and the CO₂ emissions are simultaneously minimized. This approach needs relatively less computational time since excessive operations that are required by Pareto-based techniques are avoided (ranking and pairwise comparison). To study the potential of the hybrid PV-diesel-battery system as a sustainable electrification solution for a rural Saharan community, one optimal solution was chosen among the solutions set and analyzed. The selection criteria are based on the high load growth potential in rural areas after the installation of the hybrid system and the environmental sustainability of the energy system. Then, sensitivity analysis was accomplished to study the impact of three parameters on the hybrid system (the load consumption, ϵ_{CO_2} and ϵ_{LLP} constraints). Comparative analysis between PSO based approach and HOMER methods was also performed on the chosen solution. It was observed that the PSO approach is more cost effective and results in less GHG emissions than HOMER methods. The case study focuses on the rural village of Tiberkatine, located in the province of Tamanrasset in southern Algeria at a latitude of 23° 25' N, a longitude of 6° 15'E, and an altitude of 1827 m. The village has 20 un-electrified households. The daily load is 47 kWh/day and the peak demand is 4.5 kW.

The rest of the paper is structured as follows, after the introduction, the methodology of the paper is presented. Section 3 describes the considered hybrid system and discusses the modeling of hybrid PV-diesel-battery system components, the PSO approach is presented in section 4, section 5 deals with optimization module, section 6 gives results & discussion followed by the conclusion.

2. Methodology

This paper focuses on the optimal design of a hybrid PV-diesel battery system for rural community using particle swarm optimization (PSO) approach. A comparative analysis is also performed between PSO results and HOMER results. The methodology adopted for this paper can be applied to other remote rural regions of Algeria and similar regions around the world. Our methodology includes the following steps: first, with the help of the electricity distribution company, Sonelgaz (société de distribution d'électricité du centre, SDC), our case study, the village of Tiberkatine was identified by investigating and collecting data about the number of

un-electrified households, local renewable energy resources, and meteorological characteristics of the village. Second step involved developing the domestic load profile of the Saharan community based on the standard of living, traditions, behavior, and habits. Third step consisted of mathematical modeling of the hybrid PV-diesel-battery system components. Last step involved optimization of the hybrid system using the PSO-based approach and ϵ -constraint method. Next, the energy balance of the system was derived from simulation and sensitivity analysis was established to determine the impact of three parameters (load consumption, ϵ_{CO_2} , and ϵ_{LLP} constraints) on the behavior of the optimal hybrid system followed by a comparison between HOMER software results and PSO results.

3. Modeling of hybrid PV-diesel-battery system components

The considered hybrid renewable energy system includes PV modules, storage system (Battery bank) and diesel generator. Fig. 1 illustrates the energy flow of the system. The PV modules produce electricity from solar energy to satisfy the load demand. When the energy produced is more than the needed load, the excess energy is supplied to feed the battery bank until it is full charged, when the energy produced by PV modules cannot meet the load, the batteries release energy to assist the PV modules to cover the load based on the state of charge of batteries and when the batteries are unable to meet the load requirements, the diesel generator is used as a backup unit to satisfy the load demand.

To estimate the hybrid PV-Diesel-battery system performance, different components are modelled then their combination are optimized and evaluated to meet the load demand.

3.1. PV model

We consider that the inverter includes a maximum power point tracking (MPPT) system, so the output power $P_{PV}(t)$ (W) of the PV generator during hour t of the year is calculated as shown [35]:

$$P_{PV}(t) = P_{STC} \cdot \frac{G_h(t)}{1kWh/m^2} \cdot \left[1 + \frac{\alpha}{100} (T_C(t) - 25) \right] \cdot F_{dirt} \quad (1)$$

where P_{STC} (W_p) is the output power in STC (standard test conditions), $G_h(t)$ (kWh/m^2) is the irradiation over the tilted surface of the PV panels during hour t . The angle of tilt or the slope is the

angle which the panels are mounted, in our study the tilt angle equal to the latitude of the site. F_{dirt} (around 0.9), α is the power temperature coefficient ($\%/^{\circ}C$), and $T_C(t)$ ($^{\circ}C$) is the PV cell temperature, which can be calculated as follows:

$$T_C(t) = T_a(t) + \left(\frac{NOCT - 20}{0.8} \right) \cdot \frac{G_{h_{year}}(t)}{1kWh/m^2} \quad (2)$$

where $T_a(t)$ is the ambient temperature ($^{\circ}C$) and NOCT is the nominal operation cell temperature ($^{\circ}C$).

3.2. Battery bank

The battery output current depends on the output power of the renewable generators, the AC load, the control strategy and its SOC [35]. The battery charging and discharging control with the maximum power of PV array is the key point to increase the efficiency of the generation system [36]. Energy production and absorption of battery system during the time from $t-1$ to t is described by the equation below:

$$C_B(t) = C_B(t-1)(1 - \sigma) + \left(P_{PV}(t) - \frac{P_{load}(t)}{\eta_{inv}} \right) \eta_{bat} \quad (3)$$

When:

$$P_{PV}(t) - \frac{P_{load}(t)}{\eta_{inv}} > 0 \text{ and } C_B(t-1) < C_{Bmax} \text{ the battery is charging.}$$

$$P_{PV}(t) - \frac{P_{load}(t)}{\eta_{inv}} < 0 \text{ and } C_B(t-1) > C_{Bmin} \text{ the battery is discharging.}$$

where $C_B(t)$ is the availability power of battery bank at hour t and $C_B(t-1)$ is the availability power of battery bank at the previous hour ($t-1$). The term σ is the self-discharge rate of battery bank, in the study it is assumed 0.002. $P_{load}(t)$ is the load demand at time step of t . η_{bat} is the efficiency of storage batteries ($\eta_{bat} = 0.8$) and η_{inv} is the efficiency of the inverter ($\eta_{inv} = 0.95$). The value of $C_B(t)$ is between the minimum allowable energy level remained in battery banks (C_{Bmin}), and the maximum allowable energy level (C_{Bmax}):

$$C_{Bmin} \leq C_B(t) \leq C_{Bmax} \quad (4)$$

For longevity of the battery bank, the maximum charging rate (SOC_{max}) is given as the upper limit, and it takes the value of total nominal capacity in Ah of the battery bank which is defined by the total number of batteries (N_{bat}), and the nominal capacity of each battery (C_B), as follows [37]:

$$SOC_{max} = C_B N_{bat} \quad (5)$$

The minimum permissible state of charge (SOC_{min}) of battery bank during discharging may be expressed as follows [37]:

$$SOC_{min} = N_{bat} C_B (1 - DOD_{max}) \quad (6)$$

where DOD_{max} is the maximum depth of discharge in percentage. The battery can have the maximum life if the DOD is set between 30 and 50%. In our study, $DOD = 30\%$.

Temperature can also affect the performance of batteries. High temperature decreases the battery lifetime. The batteries are normally installed inside a building where the temperature is not expected to change drastically. It is recommended to keep the battery at $25^{\circ}C$ [10].

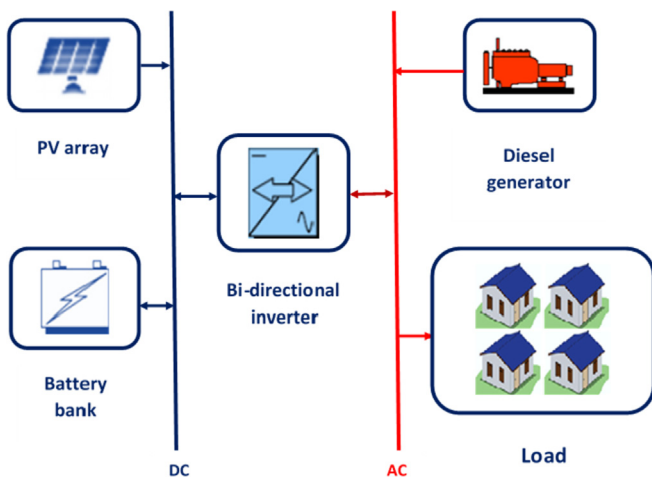


Fig. 1. Energy flow of the hybrid system.

3.3. Inverter

The power transiting the inverter to serve the demand is given by:

$$P_{in} = \frac{P_{ch}}{\eta_{in}} \quad (7)$$

η_{inv} is the inverter efficiency specified by the manufacturer (%), P_{ch} is the hourly demand (W).

3.4. Diesel generator fuel consumption

The fuel consumption of the diesel generator, $Cons_G$ (L/h) is modelled as dependent on the output power [38]:

$$Cons_G = \alpha_{DG} P_{n-DG} + \beta_{DG} P_{a-DG} \quad (8)$$

where P_{n-DG} (kW) is the nominal power, P_{a-DG} (kW) is the power output of the diesel generator, α_{DG} and β_{DG} are the coefficients of the consumption curve, defined by the user (L/kWh). The diesel efficiency η_G is defined as power output divide by heating value of fuel consumption (LHV_{fuel}) calculated using the following equation:

$$\eta_G = \frac{P_{a-DG}}{Cons_G LHV_{fuel}} \quad (9)$$

where LHV_{fuel} is between 10 and 11.6 kWh/L, $\alpha_{DG} = 0.246$ L/kWh and $\beta_{DG} = 0.08145$ L/kWh [48].

4. PSO based approach

To find the optimal sizing of hybrid PV-diesel-battery system using PSO and ϵ -constraint, three objective functions are considered (the total cost of the system, the total CO₂ emissions produced by diesel generator, and the loss of load probability LPP). Design variables included in the optimization process are: the size of the PV panels N_{pv} , the size of the battery bank N_{bat} , and the rated capacity of DG unit P_{n-DG} . These variables are defined in a vector named particle $x_i = [N_{pvi}, N_{bati}, P_{n-DGi}]$. Each particle represents different configuration of the system. Total cost, is a function of design variables and is considered as the fitness of particles for particles evaluation while CO₂ emission and unmet load are considered as constraints bounds. The optimization and simulation flowchart using PSO and ϵ -constraint method is shown in Fig. 2. After entering the initial parameters and different constraints of the system such as the hourly load demand, hourly meteorological data, lifetime of the project, cost of the components etc., the next step is the initializing of the first population of particles. Particles are randomly generated then in each iteration and for each particle, the energy flow between components is evaluated, the LLP and CO₂ emission are calculated. The results of the simulation procedure are sent back to the optimization model to calculate the fitness value for each particle and check its feasibility (if it meets the constraints of the model).

The simulation model is run for one year to evaluate the performance of each particle and is carried out based on weather data (insolation, temperature, etc.). Simulation model calculates the annual fuel consumption of diesel generator unit, the CO₂ emission and the yearly-unmet load for each particle. These values are sent to the optimization algorithm to check if the particle meets the constraints (the desirable level of unmet load and CO₂ emission). If the particle did not meet the constraints, it is modified and sent back to simulation. In each iteration, feasible particles are evaluated in the PSO algorithm based on their fitness until termination

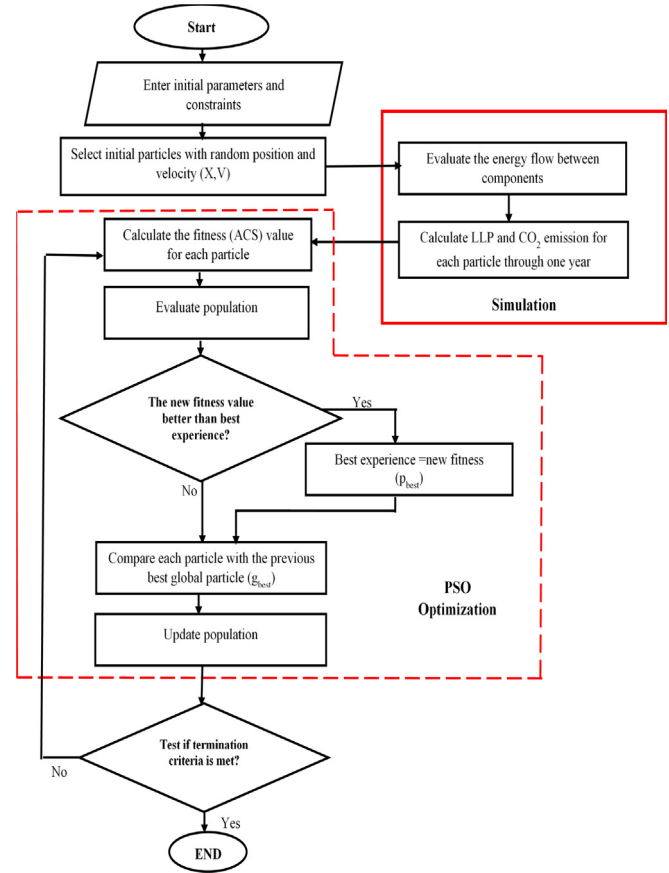


Fig. 2. Simulation and optimization flowchart of the hybrid PV-diesel-battery system.

criterion is checked. After terminating the cycle, the optimal solution will be returned.

5. Optimization module and objective functions

5.1. Particle swarm optimization algorithm

PSO is a meta-heuristic optimization technique. PSO is inspired firstly by general artificial life, the same as bird flocking, fish schooling and social interaction behavior of human and secondly by random search methods of the evolutionary algorithm [39].

It does not require good initial solutions to start its iteration process. The particle moves towards an optimum solution through its present velocity and its individual best solution obtained by itself in each iteration and global best solution is obtained by all particles. In a physical dimensional search space, updated position and velocity of the 'ith' particle are represented as:

$$v_i^{k+1} = K[v_i^k * \omega + C_1 * R_1 * \{p_{best}(i) - x_i^k\} + C_2 * R_2 * \{g_{best} - x_i^k\}] \quad (10)$$

$$x_i^{k+1} = x_i^k + v_i^{k+1} \quad (11)$$

where:

$$x_i = [x_{i1}, x_{i2}, \dots, x_{id}]$$

$$v_i = [v_{i1}, v_{i2}, \dots, v_{id}]$$

1, 2, ..., d shows possible dimension for $i = 1, 2, \dots, n$ particles with position x and velocity v

$p_{best}(i) = [X_{i1p_{best}}, X_{i2p_{best}}, \dots, X_{idp_{best}}] p_{best}$ is the individual best position of particle i

$g_{best} = [X_{1G_{best}}, X_{2G_{best}}, \dots, X_{nG_{best}}]$ is the global best position, k represents the iteration number for total n iterations.

K is constriction factor, C_1 and C_2 are non-negative coefficients called acceleration factors, R_1 and R_2 are two random numbers different from each other and generally distributed in the range $[0,1]$. ω is inertia weight, suitable selection of ω provides a balance between global exploration and local exploitation reduces the total number of iterations [40].

5.2. The ε -constraint method

The ε -constraint method is a simple MOP (Multi-Objective optimization) technique that can be used where one objective is chosen to be optimized. The remaining objectives are considered as constraints bounded by given target levels (ε_i) [41]. By varying these levels, the non-inferior solutions of introduced problem can be obtained. Consider the following MOP:

$$\text{Min} \{f_1(x), f_2(x), \dots, f_k(x)\} \quad (12)$$

where x is the decision vector, f_i ($i = 1, 2, \dots, k$) are objective functions.

A solution x^* is said to be non-inferior if there exists no other feasible solution x such that $f_i(x) \leq f_i(x^*)$ for all $i = 1, 2, \dots, k$, and at least, one inequality is strict.

In the ε -constraint method, if $f_j(x)$, $j \in \{1, \dots, k\}$ is the objective function chosen to be optimized, and $f_i(x)$ is objective considered as the constraint, we have the following problem [41,42].

$$\begin{aligned} \text{Min } f_j(x) \quad & j \in \{1, \dots, k\} \text{ subject to } f_i(x) \\ & \leq \varepsilon_i \quad \forall i \in \{1, \dots, k\}, i \neq j, x \in S \end{aligned} \quad (13)$$

where S is the feasible solution space, k is the number of objective functions, and ε_i is assumed values of the objective function that must not be exceeded.

5.3. Objective function and constraints

The annual cost of the system (ACS) is considered as objective function and CO_2 emission and LLP during one year are considered as the constraint bounds.

The system costs consist of: annual capital cost (ACC), annual operation and maintenance cost (AOM), annual fuel cost (AFC), and annual replacement cost (ARC), ACS is calculated using the following equation:

$$\text{ACS} = \text{ACC} + \text{AOM} + \text{ARC} + \text{AFC} \quad (14)$$

Annual capital cost of each unit that does not need replacement during project lifetime such as DG, PV, and inverter is calculated as follows:

$$\text{ACC} = C_{cap} \text{CRF}(i, y) \quad (15)$$

where C_{cap} is the capital cost of each component in US\$, y is the project lifetime in a year. CRF is capital recovery factor, a ratio to calculate the present value of a series of equal annual cash flows. This factor is calculated as follows:

$$\text{CRF} = \frac{i(1+i)^y}{(1+i)^y - 1} \quad (16)$$

where i is the annual real interest rate. Annual real interest rate consists of nominal interest i' and annual inflation rates f . This rate is calculated as follow:

$$i = \frac{(i' - f)}{(1 + f)} \quad (17)$$

Annual operation and maintenance cost of system (AOM) can be determined using the following equation [43,44]:

$$\text{AOM} = \text{AOM}(1) (1 + f)^y \quad (18)$$

where $\text{AOM}(1)$ is the maintenance cost of that component for the first year of the project.

ARC is annual cost value for replacing battery banks during the project lifetime. Economically, annual replacement cost is calculated using the following equation [45]:

$$\text{ARC} = C_{rep} \text{SFF}(i, y_{rep}) \quad (19)$$

where C_{rep} is the replacement cost of battery banks in US\$, y_{rep} is the lifetime of battery banks in a year. SFF is the sinking fund factor, a ratio to calculate the future value of a series of equal annual cash flows. This factor is calculated as following [39]:

$$\text{SFF} = \frac{i}{(1+i)^y - 1} \quad (20)$$

The economic parameters of the system are presented in Table 1.

5.4. The problem constraints

As previously mentioned, the problem constraints of our system are Loss of load probability of the system LLP and CO_2 emissions:

LLP is the probability that a generation will be insufficient to meet demand at some point over some specific time window [46]. LLP is defined by the following equation, and it should be less than allowable LLP reliability index: [38,47]

$$\text{LLP} = \frac{\sum_{t=1}^{8760} \text{shortage}(t)}{\sum_{t=1}^{8760} D(t)} \quad (21)$$

where, $\text{LLP} \leq \varepsilon_{\text{LLP}}$ and $D(t)$ is electricity demand, and $\text{shortage}(t)$ is unmet load during time period t [38].

The CO_2 quantity produced (in kg) by diesel generator is considered to represent the pollutant emission and is calculated using Eq. (22) [38,47]. The CO_2 emission should be less than allowable emission level:

$$\text{CO}_{2\text{emission}} = \sum_{t=1}^{8760} \text{fuel}_{\text{cons}}(t) \text{EF} \quad (22)$$

Table 1
Economic parameters of the hybrid PV diesel system.

Parameters	Value
Annual interest rate i (%)	8.25
Inflation rate f (%)	8.17
Project lifetime (years)	20
MPPT efficiency (%)	96
PV panel lifetime (years)	20
Battery banks lifetime (years)	7
Inverter lifetime (years)	20
Reliability of PV modules	0.98
Reliability of inverter	0.95
Reliability of battery banks	0.9
Reliability of diesel generator	0.9
Cost of PV panel (\$/W)	2.3
Cost of diesel generator (\$/kW)	375
Cost of a battery (\$/kWh)	400
Cost of an inverter (\$/kW)	711
Fuel cost (\$/L)	0.16
Emission factor (kg/L)	2.4

where $CO_{2_{emission}} \leq \varepsilon_{CO_2}$ and EF is the emission factor for a diesel generator. It depends on the type of fuel and diesel engine characteristics. It is considered between 2.4 kg/L and 2.8 kg/L [48].

Some other feasible operation constraints should be considered:

- The constraints of the number of PV modules and batteries and capacity of DG:

$$N_{PV}, N_{bat}, P_{n-DG} \geq 0 \quad (23)$$

- The constraints of the capacity of batteries:

$$C_{Bmin} \leq C_B(t) \leq C_{Bmax} \quad (24)$$

-The energy flow (energy produced by or entered to each component) in every time ($E_j(t)$) should be less than the capacity of the component, where Δt is the time interval ($\Delta t = 1$ h).

$$E_j(t) \leq P_j \Delta t \quad (25)$$

6. Results and discussion

In this paper, the hourly behavior simulation of the hybrid components is calculated under a load following strategy in both approaches (PSO approach and HOMER). Under the Load Following strategy, when the generator operates, it produces only enough power to meet the primary load. Charging the battery bank or serving the deferrable load are left to the renewable power sources. Load following tends to be optimal in systems with a lot of renewable power that sometimes exceeds the load.

6.1. Presentation of the site

The hybrid PV-diesel-battery system is designed to electrify 20 households located in the rural Saharan village namely Tiberkatine, the village is 2 km away from Tazrouk district, Tamanrasset province, in the southern Algeria (latitude $23^\circ 25'$ N and longitude $6^\circ 15'E$).

6.2. Load profile

Energy demand of our village is classified as domestic load. The load profile incorporates the lighting with CFL (Compact Fluorescent Lamp), TV, fan, radio for each household (basic household load). Two daily load profiles are represented (winter and summer daily load profile), the constant sequence of power over a step of

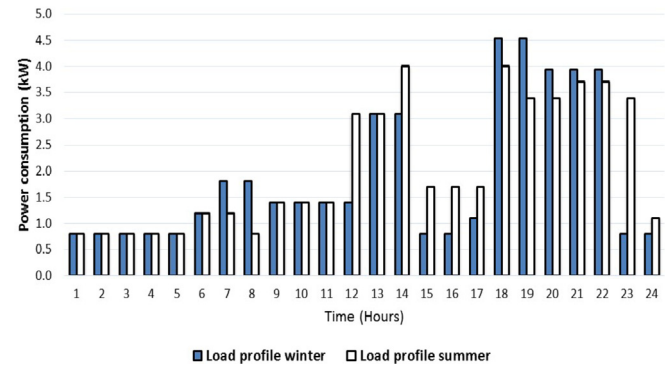


Fig. 3. Load profiles variations of the remote village Tiberkatine (province of Tamanrasset, Algeria).

time is 1 h as shown in Fig. 3. Suggested load profiles are derived based on standard of living, traditions, behavior and habits. The daily load is 47 kWh/day and peak demand is around 4.5 kW.

6.3. Meteorological data

In remote inaccessible sites, weather data collection is very difficult, therefore, we were forced to use the synthetic hourly solar data generated by HOMER from monthly average data and sourced from climate database of the NASA Surface Meteorological dataset [49]. The hourly solar radiation data are synthesized using an algorithm based on the work of V.A. Graham [50]. The other set of data containing the ambient temperature is obtained from measured data of the year 1992 [51]. Fig. 4 shows the meteorological characteristics (ambient temperature, and daily radiation) of the site. One-year meteorological data (insolation and temperature) are considered. Tiberkatine has hot desert climate, with hot summers and cool winters, and slight annual precipitation. The climate is quite moderate compared to the rest of the Sahara region due to high altitude, and precipitation.

6.4. PSO simulation results

The Program has been coded in MATLAB. The swarm size of PSO algorithm consist of 40 particles, the particle is a vector of three

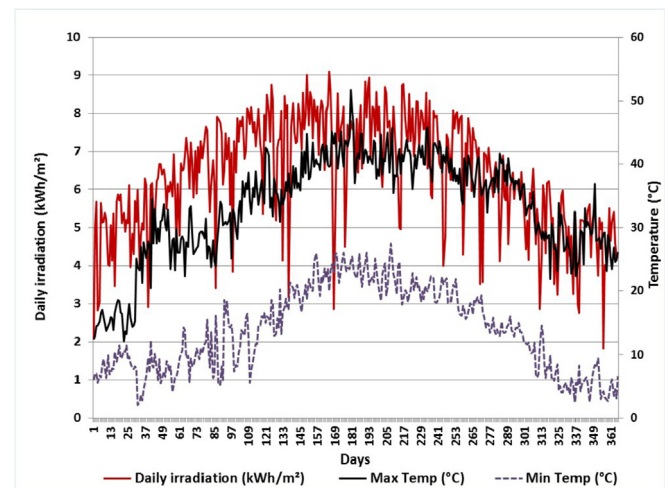


Fig. 4. Meteorological data of the site village of Tiberkatine (province of Tamanrasset, Algeria).

elements [the size of PV panels, the number of batteries, and the capacity of diesel generator]. Each vector represents certain configuration of the system. The considerable size for PV is $N_{pv} \in [10-95]$, for battery bank is $N_{bat} \in [2-48]$ and for diesel generator is $P_{n-DG} \in [5-10]$ (kW).

The best interval for PSO parameters ω , C_1 , C_2 , R_1 and R_2 are suggested according to literature: acceleration coefficients C_1 and C_2 are set within the range of $[0-4]$, R_1 and R_2 are random numbers set within range of $[0-1]$. In this study, we use the formula suggested to calculate the inertia weight (ω) [52]:

$$\omega(j) = \omega_0 \exp\left(\frac{j}{\max j}\right)^n \quad (26)$$

where ω_0 is initial weight, j is iteration number, $\max j$ is maximum iteration number and n is curve shape control power.

The goal of the PSO and ϵ -constraint approach is to minimize one objective function at a time which is the ACS of the system, and considering the other objectives as constraints bound (LLP and CO₂ emissions) by some allowable levels (ϵ_{CO_2} , ϵ_{LLP}). By varying these levels, the non-inferior solutions of the problem can be obtained. Preliminary analysis is required to identify proper starting values for ϵ_{CO_2} and ϵ_{LLP} . To design a sustainable electrification solution for a rural Saharan community, one optimal solution was chosen after the optimization phase. The selection criteria are based on the high load growth potential in rural areas after the installation of the hybrid system and the environmental sustainability of the energy system, therefore, to handle the growth of the load demand and to avoid the maximum GHG emissions, in our case, the optimal solution with $\epsilon_{CO_2} \leq 2000$ kg/year, $\epsilon_{LLP} \leq 0.5\%$ is picked. The optimal size of the components is 11.4 kWp of PV generator, 42 kWh of battery bank and 6 kW of diesel generator.

ACS of the optimal system is 8585.14 \$, COE is 0.38 \$/kWh, the cost of the battery bank is the dominant cost of the system with 51.4% of the ACS. The economical results of the system are reported in Table 2.

An example of simulating intra-hourly energy fluctuations of the hybrid system in winter day (the 3rd of January) and summer day (23rd of July) is represented in Fig. 5: when the energy produced by the PV generator is more than the load requirement, the load is satisfied by the PV subsystem. When the energy produced by PV generator cannot meet the load, the battery bank assists the PV subsystem to cover the load based on the state of charge of batteries. Otherwise, the diesel generator is used as a backup unit to satisfy the load demand. Local weather variation (mainly the solar radiation) has direct influence on the operation and the energy contribution of the diesel generator, and it is clearly shown in winter day.

The size of the components is enough to satisfy the load demand without any unmet load. Monthly variation of the energy production from PV, diesel generator and the storage capacity of the battery bank are presented in Fig. 6. PV production has a significant impact on the storage capacity and the diesel production, the PV production and the storage capacity of the battery bank are linearly related, when the PV production is favorable (months of October

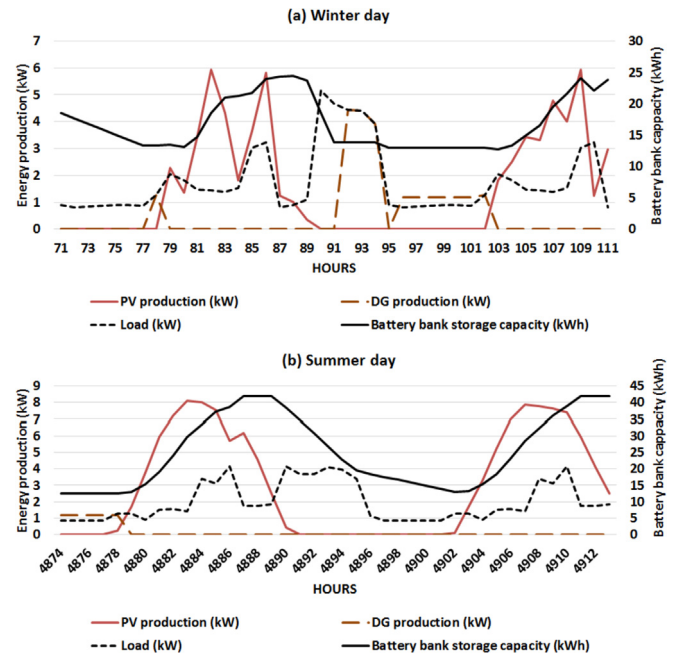


Fig. 5. Simulating intra-hourly energy fluctuations of the hybrid system.

and March), the storage capacity is maximal, and when the PV production decreased, the storage capacity also decreased. Unlike the storage capacity, the PV production and the DG production are inversely related because the DG is used as back up unit when the PV subsystem cannot meet the load requirements.

6.5. Sensitivity analysis

Sensitivity analysis is performed to investigate the impact of the variation of the input parameters on the behavior of the system and to identify the most sensitive parameters. In this paper, the parameters are: the load consumption, ϵ_{CO_2} constraint and ϵ_{LLP} constraint.

6.5.1. Load consumption

The load consumption is varied from 49.4 kWh/d to 89.4 kWh/d, the sizes of PV, battery bank, and diesel generator are remained the same. To cover the capacity shortage due to the growth of the load, the system relies mainly on the diesel generator. As shown in Fig. 7 and Table 3, the dependence on the diesel subsystem increases the ACS of the system from 8585 \$ to 9347 \$ (an increase of 8.8%) due to the increasing costs of the fuel consumption. It is also observed that the COE decreases from 0.38 \$/kWh to 0.26 \$/kWh due to the

Table 2
The economic results of the optimal size.

	ACC (US\$)	AOM (US\$)	ARC (US\$)	AFC (US\$)	ACS (US\$)
PV subsystem	2704.7	126.19	0	0	2830.89
Diesel generator	231.96	108.22	0	133	473.18
battery bank	1731.9	808.07	1873.4	0	4413.37
converter	439.89	205.19	222.63	0	867.71
Total	5108.45	1247.67	2096.07	133	8585.14

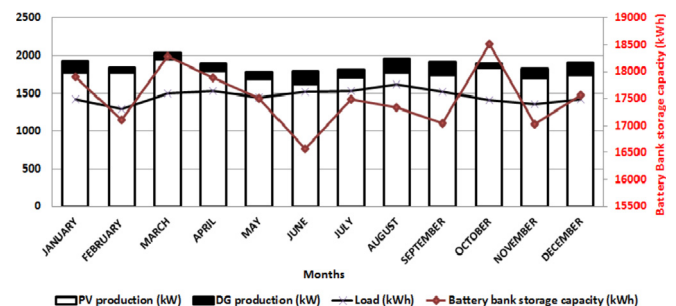


Fig. 6. The monthly variation of the energy production from PV, diesel generator and battery bank storage.

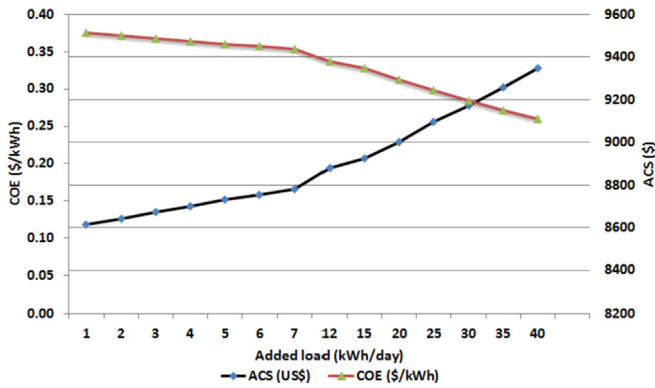
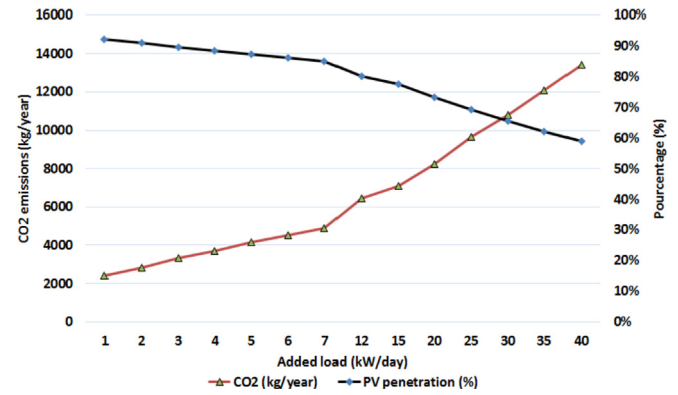


Fig. 7. Impact of load variation on the NPC and COE of the system.

Fig. 8. Effect of load variation on the PV penetration and CO₂ emission.

decreasing of the PV penetration from 93% to 59% (see Fig. 8), which means that the fuel cost influences the COE more than the cost of the PV (the cost of the conventional energy is less than the cost of the PV energy). Subsequently, the CO₂ emissions increases from 1994 kg/yr to 13428 kg/yr as shown in Fig. 8. and the LLP increases from 0% to 7.46%.

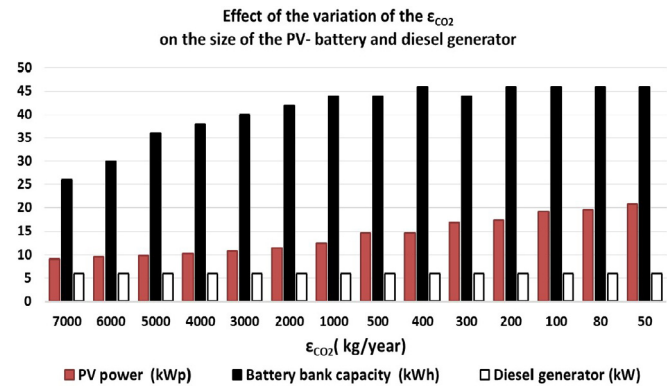
6.5.2. ϵ_{CO_2} and ϵ_{LLP} constraints

The Fig. 9 depicts the effect of ϵ_{CO_2} on the size of the PV, battery and diesel generator. ϵ_{CO_2} has a significant impact on the size of the system basically the size of the battery bank, the size of battery bank increases from 26 kWh to 46 kWh when ϵ_{CO_2} ranges from 7000 kg/yr to 50 kg/yr. Whereas for the effect of ϵ_{LLP} on the size of the system, the size of the battery bank is also the most influenced, it decreases from 22 kWh to 18 kWh when ϵ_{LLP} ranges from 0% to 2% as shown in Fig. 10.

The Fig. 11 shows the effect of the variation of ϵ_{CO_2} on the PV penetration, the PV penetration is inversely related to ϵ_{CO_2} , it increases from 74.5% to 99.9% when ϵ_{CO_2} is varied. As for the effect of the variation of ϵ_{LLP} , the PV penetration decreases slightly from 72% to 67% as depicted in Fig. 12.

The Fig. 13 shows the effect of the variation of ϵ_{LLP} on different costs. ACS is inversely related to ϵ_{LLP} of the system, ACS decreases by 717.4 \$. Furthermore, COE ranges between 0.24 \$/kWh and 0.27 \$/kWh when ϵ_{LLP} is varied.

Whereas for the effect of the variation of ϵ_{CO_2} on different costs, ACS is also inversely related to ϵ_{CO_2} of the system as seen in Fig. 14, ACS increases from 6662.8 \$ to 11231.3 \$ (an increase of 4568.4 \$). As for the impact of ϵ_{CO_2} on COE, it can be observed that COE ranges between 0.29 \$/kWh and 0.38 \$/kWh.

Fig. 9. Effect of ϵ_{CO_2} on the size of the PV, battery and diesel generator.

6.6. System simulation with HOMER

Hybrid optimization model for electric renewables software (HOMER) is used to simulate and optimize the hybrid system of our case study. The software, NREL's micropower optimization model, can evaluate a range of equipment options over varying constraints and sensitivities to optimize small power systems [53]. It is used to identify the cost-minimizing combination of power generating technologies that would suit the load profiles. Therefore, the software gives us the opportunity to design a cost-minimizing power system that provides a tailor-made power supply for the specific load demand [54].

The following equation is used to calculate the total net present

Table 3
Effect of varying the load consumption on PV penetration, LLP, CO₂ emission, ACS and COE.

The added load (kWh/day)	PV penetration (%)	LLP (%)	CO ₂ (kg/year)	ACS (US\$)	COE (\$/kWh)
1	92.15	0	2429.8	8614.1	0.38
2	90.97	0	2818.9	8640.1	0.37
3	89.58	0	3326.9	8673.9	0.37
4	88.43	0	3717.6	8700	0.36
5	87.21	0	4156.9	8729.3	0.36
6	86.13	0	4530.2	8754.1	0.36
7	85.09	0	4895.7	8778.5	0.35
12	80.24	0	6427.9	8880.7	0.34
15	77.62	0	7070	8923.5	0.33
20	73.30	0.37	8225	9000.5	0.31
25	69.19	1.67	9645.6	9095.2	0.30
30	65.41	3.71	10788	9171.3	0.28
35	61.95	5.62	12078	9257.3	0.27
40	58.80	7.46	13428	9347.3	0.26

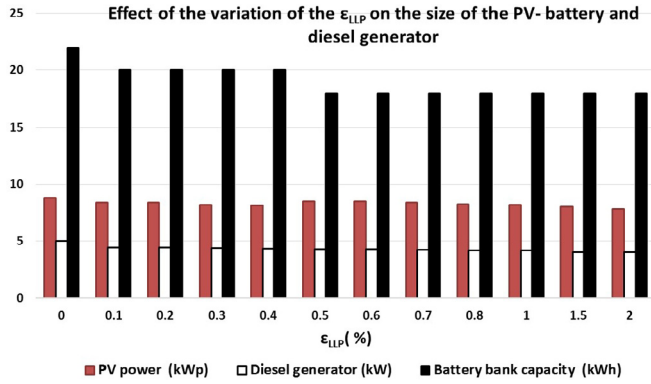


Fig. 10. Effect of ϵ_{LLP} on the size of the PV, battery and diesel generator.

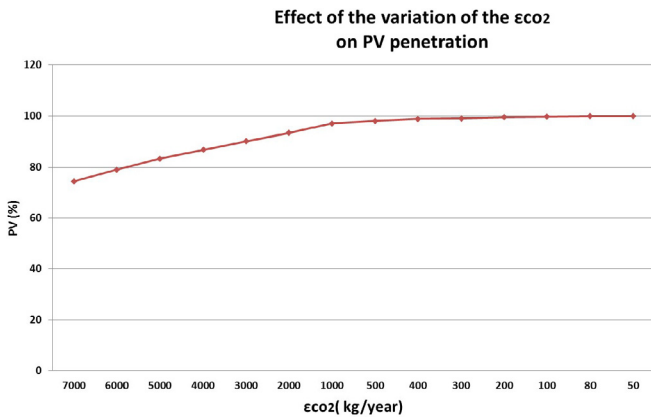


Fig. 11. Effect of the variation of ϵ_{CO2} on the PV penetration.

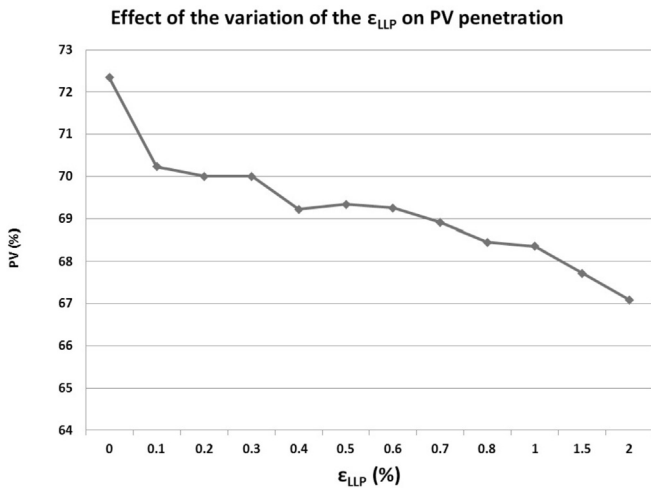


Fig. 12. Effect of the variation of ϵ_{LLP} on the PV penetration.

cost of the system [55]:

$$C_{NPC} = \frac{C_{ann,tot}}{CRF(i, R_{proj})} \quad (27)$$

where $C_{ann,tot}$ is the total annualized cost, i the annual real interest rate (the discount rate), R_{proj} the project lifetime, and CRF is the capital recovery factor, given by the following equation:

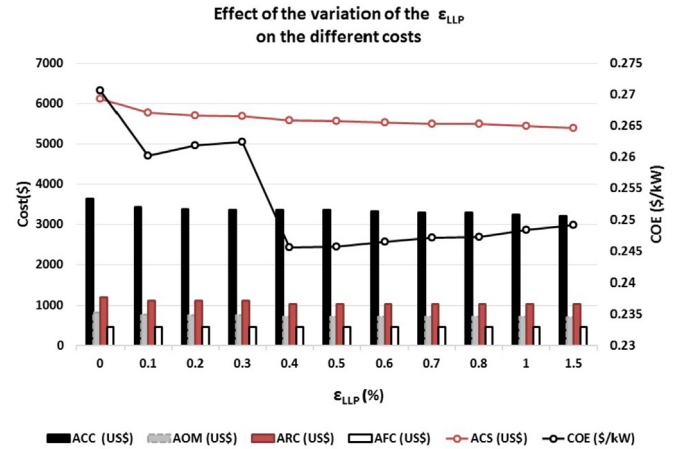


Fig. 13. Effect of the variation of ϵ_{LLP} on the different costs.

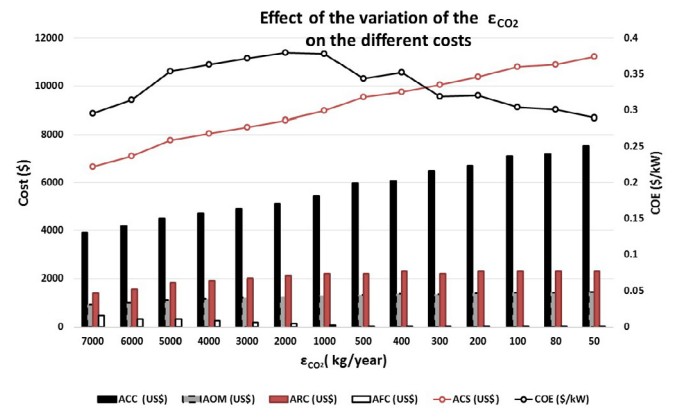


Fig. 14. Effect of the variation of ϵ_{CO2} on the different costs.

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (28)$$

where i is the annual real interest rate and N is the number of years.

The equation of energy levelized cost is given as follows:

$$COE = \frac{C_{ann,tot}}{E_{prim} + E_{def} + E_{grid,sales}} \quad (29)$$

where $C_{ann,tot}$ is the total annualized cost, E_{prim} is the total amount

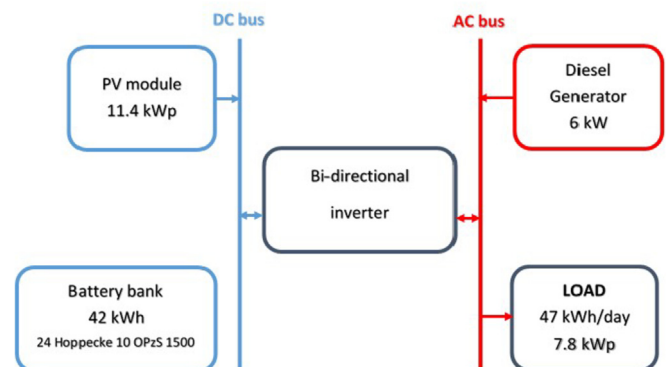


Fig. 15. The schematic of the hybrid system.

Table 4

The economic results of the system.

Net Present Costs						
Component	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)	Total (\$)
PV	26220	0	2527	0	0	28747
Generator 1	2250	0	1999	1227	–107	5368
Hoppecke 10 OPzS 1500	16800	9670	1349	0	–1011	26809
Converter	4266	1738	463	0	–307	6159
System	49536	11408	6338	1227	–1425	67083

Table 5

The simulation results from PSO and HOMER.

	PSO	HOMER
PV array production (kWh/yr)	22134 (93%)	21056 (91%)
Diesel generator (kWh/yr)	1498 (7%)	2076 (9%)
Total (kWh/yr)	22633	23132
Fuel consumption (L/yr)	831	872
energy input of batteries (kWh/yr)	9687	9966
energy output of batteries (kWh/yr)	9711	8586
CO ₂ emissions (kg/yr)	1994	2094

of primary load serves per year, E_{def} is the total amount of deferable load serves per year and $E_{\text{grid, sales}}$ is the amount of energy sold to the grid per year.

To calculate the salvage value of each component at the end of the project lifetime, the following equation is used:

$$S = C_{\text{rep}} \cdot \frac{R_{\text{rem}}}{R_{\text{comp}}} \quad (30)$$

where S is the salvage value, C_{rep} is the replacement cost of the component, R_{rem} is the remaining life of the component, and R_{comp} is the lifetime of the component.

6.6.1. Results and comparison

The hourly behavior simulation of the hybrid components is calculated under “the load following dispatch strategy”: the load is supplied by the PV system and the diesel generator supplies the load only when the batteries are discharged. The schematic of the used hybrid system is illustrated in Fig. 15. The different.

The economic results of the system from HOMER are reported in Table 4; the initial capital cost of system is 49536 \$ represents 73.8% of the total NPC (net present cost). The cost of the PV subsystem is the dominant cost (42%) followed by the battery bank (40%). The cost of energy (COE) is 0.4 \$/kWh.

The simulation results from PSO and HOMER are reported in Table 5. The comparative analysis shows that the optimal hybrid system of PSO is more cost effective than HOMER system. The cost of energy is 0.38 \$/kWh for PSO and 0.4 \$/kWh for HOMER. It is also found that PSO optimal system has more renewable penetration than HOMER system, the PV penetration is 91% for HOMER and 93% for PSO. Furthermore, The CO₂ gas emission given by PSO optimal system is 1994.5 kg/year and for HOMER optimal system is 2094 kg/year.

The hybrid system supplies the load consumption without an unmet load in both approaches. However, PSO approach is more cost effective with more renewable penetration than HOMER, and it has also less CO₂ emission than HOMER.

7. Conclusion

An optimization model based on PSO and ϵ -constraint method is effectively established to search for the optimal design of an off-

grid hybrid PV-diesel-battery system as rural reliable electrification solution. Three objectives functions are considered (annualized cost of the system ACS, loss of load probability LLP, and CO₂ emissions). Hybridizing PSO with other algorithms can complicate the system, but in our case, the ϵ -constraint method is used to handle constraints and multiple objectives of the system with simplicity and computational efficiency. The results indicate that the optimal system satisfies 100% of the load demand, the COE of the optimal system is 0.38 \$/kWh, the PV penetration is 93%, the CO₂ emission is 1994 kg/year and the dominant cost of the system is the cost of the battery bank (51%) plus the PV subsystem (33%). A sensitivity analysis is performed on three parameters shows that the load consumption growth has direct impact on reducing the PV penetration, whereas, the diesel generator satisfies the supplementary load. The fuel consumption, the CO₂ emission and the O&M cost also increase, which leads to increase the ACS of the system. Moreover, The CO₂ emissions constraint ϵ_{CO_2} strongly affect the cost of the system, the size of the battery bank and the size of the PV panels. Decreasing the ϵ_{CO_2} has increased the ACS, PV penetration and the size of the PV panels, and the capacity storage of the battery bank. It is also found that decreasing the ϵ_{LLP} has increased the ACS and the size of the PV subsystem and the battery bank. Furthermore, the comparison between PSO based optimization and HOMER results show that PSO based approach is more cost effective and has quite high renewable penetration with low value of CO₂ emissions compared to HOMER results. The future work will investigate the optimization of other combination of renewable sources with different control strategies in the designing process of HRES for rural and Saharan communities.

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